BASIC INTERVIEW LEVEL

1️⃣ Question:

Write a program to check whether a number is even or odd.

```
n = int(input())
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

2️⃣ Question:

Write a program to check whether a number is positive, negative, or zero.

```
n = int(input())
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

3️⃣ Question:

Write a program to find the largest among three numbers without using max().

```
a=int(input()); b=int(input()); c=int(input())
if a>=b and a>=c:
    print("Largest:",a)
elif b>=a and b>=c:
    print("Largest:",b)
else:
    print("Largest:",c)
```

4️⃣ Question:

Write a program to find the smallest among three numbers.

```
a=int(input()); b=int(input()); c=int(input())
if a<=b and a<=c:
    print("Smallest:",a)
elif b<=a and b<=c:
    print("Smallest:",b)
else:
    print("Smallest:",c)
```

5️⃣ Question:

Write a program to check whether a year is a leap year.

```
y=int(input())
if (y%4==0 and y%100!=0) or y%400==0:
    print("Leap Year")
else:
    print("Not Leap Year")
```

6️⃣ Question:

Write a program to check whether a character is a vowel or consonant.

```
ch=input()
if ch=='a' or ch=='e' or ch=='i' or ch=='o' or ch=='u' or \
   ch=='A' or ch=='E' or ch=='I' or ch=='O' or ch=='U':
    print("Vowel")
else:
    print("Consonant")
```

7️⃣ Question:

Check whether a number is divisible by both 3 and 5.

```
n=int(input())
if n%3==0 and n%5==0:
    print("Divisible")
else:
    print("Not Divisible")
```

8️⃣ Question:

Check if a number is a two-digit number.

```
n=int(input())
if (n>=10 and n<=99) or (n<=-10 and n>=-99):
    print("Two Digit")
else:
    print("Not Two Digit")
```

9️⃣ Question:

Determine whether a person is eligible to vote.

```
age=int(input())
if age>=18:
    print("Eligible")
else:
    print("Not Eligible")
```

🔟 Question:

Assign grade based on marks using if-elif ladder.

```
m=int(input())
if m>=90:
    print("A")
elif m>=80:
    print("B")
elif m>=70:
    print("C")
elif m>=40:
    print("Pass")
else:
    print("Fail")
```

INTERMEDIATE INTERVIEW LEVEL

11 Question:

Check whether three sides form a valid triangle.

```
a=int(input()); b=int(input()); c=int(input())
if a+b>c and a+c>b and b+c>a:
    print("Valid Triangle")
else:
    print("Invalid Triangle")
```

12 Question:

Determine the type of triangle.

```
a=int(input()); b=int(input()); c=int(input())
if a==b and b==c:
    print("Equilateral")
elif a==b or b==c or a==c:
    print("Isosceles")
else:
    print("Scalene")
```

13 Question:

Check whether a number is prime.

```
n=int(input())
if n<=1:
    print("Not Prime")
else:
    count=0
    for i in range(1,n+1):
        if n%i==0:
            count+=1
    if count==2:
```

```
        print("Prime")
    else:
        print("Not Prime")
```

14 Question:

Check whether a number is a perfect square.

```
n=int(input())
i=1
flag=0
while i*i<=n:
    if i*i==n:
        flag=1
        break
    i+=1

if flag==1:
    print("Perfect Square")
else:
    print("Not Perfect Square")
```

15 Question:

Find the second largest among three numbers.

```
a=int(input()); b=int(input()); c=int(input())
if (a>=b and a<=c) or (a<=b and a>=c):
    print("Second:",a)
elif (b>=a and b<=c) or (b<=a and b>=c):
    print("Second:",b)
else:
    print("Second:",c)
```

16 Question:

Implement a simple calculator.

```
a=int(input()); b=int(input()); op=input()
if op=='+':
    print(a+b)
elif op=='-':
    print(a-b)
elif op=='*':
    print(a*b)
elif op=='/':
    if b!=0:
        print(a/b)
    else:
        print("Division by Zero")
```

17⃣ Question:

Check whether a number is palindrome (without string).

```
n=int(input())
temp=n
rev=0
while n>0:
    d=n%10
    rev=rev*10+d
    n//=10
if temp==rev:
    print("Palindrome")
else:
    print("Not Palindrome")
```

18⃣ Question:

Check whether a 3-digit number is Armstrong.

```
n=int(input())
temp=n
s=0
while n>0:
    d=n%10
    s+=d*d*d
    n//=10
if s==temp:
    print("Armstrong")
else:
    print("Not Armstrong")
```

INTERMEDIATE (CONTINUED)

19⃣ Question:

Calculate electricity bill based on slabs:

First 100 units → ₹5/unit

Next 100 units → ₹7/unit

Above 200 units → ₹10/unit

```
unit=int(input())
if unit<=100:
    bill=unit*5
elif unit<=200:
    bill=100*5+(unit-100)*7
```

```
else:
    bill=100*5+100*7+(unit-200)*10
print("Bill:",bill)
```

20 Question:

Validate a strong password (must contain uppercase, lowercase, digit, special character).

```
p=input()
u=l=d=s=0
for ch in p:
    if ch>='A' and ch<='Z':
        u=1
    elif ch>='a' and ch<='z':
        l=1
    elif ch>='0' and ch<='9':
        d=1
    else:
        s=1

if u and l and d and s:
    print("Strong Password")
else:
    print("Weak Password")
```

ADVANCED INTERVIEW LEVEL

21 Question:

Check whether a year is a century leap year (400 rule).

```
y=int(input())
if y%400==0:
    print("Leap Year")
elif y%100==0:
    print("Not Leap Year")
elif y%4==0:
    print("Leap Year")
else:
    print("Not Leap Year")
```

22 Question:

Determine profit or loss percentage.

```
cp=float(input())
sp=float(input())

if sp>cp:
    profit=sp-cp
```

```
    print("Profit %:",(profit/cp)*100)
elif cp>sp:
    loss=cp-sp
    print("Loss %:",(loss/cp)*100)
else:
    print("No Profit No Loss")
```

23 Question:

Solve quadratic equation and classify roots.

```
a=int(input()); b=int(input()); c=int(input())
d=b*b-4*a*c

if d>0:
    print("Real and Distinct")
elif d==0:
    print("Real and Equal")
else:
    print("Complex Roots")
```

24 Question:

Implement ATM withdrawal system (minimum balance ₹1000).

```
balance=int(input())
withdraw=int(input())

if withdraw<=balance-1000:
    balance=balance-withdraw
    print("Withdraw Successful")
    print("Remaining:",balance)
else:
    print("Insufficient Balance")
```

25 Question:

Create loan eligibility system (salary ≥ 25000, age ≥ 21, credit score ≥ 650).

```
salary=int(input())
age=int(input())
credit=int(input())

if salary>=25000 and age>=21 and credit>=650:
    print("Eligible")
else:
    print("Not Eligible")
```

26 Question:

Determine BMI category.

```
weight=float(input())
height=float(input())
bmi=weight/(height*height)

if bmi<18.5:
    print("Underweight")
elif bmi<25:
    print("Normal")
elif bmi<30:
    print("Overweight")
else:
    print("Obese")
```

27⃣ Question:

Create grading system with subject pass condition (pass ≥ 40 in all).

```
m1=int(input()); m2=int(input()); m3=int(input())
avg=(m1+m2+m3)/3

if m1>=40 and m2>=40 and m3>=40:
    if avg>=75:
        print("Distinction")
    else:
        print("Pass")
else:
    print("Fail")
```

28⃣ Question:

Check whether three numbers form right-angled triangle.

```
a=int(input()); b=int(input()); c=int(input())

if a*a+b*b==c*c or a*a+c*c==b*b or b*b+c*c==a*a:
    print("Right Triangle")
else:
    print("Not Right Triangle")
```

29⃣ Question:

Menu-driven banking system.

```
balance=10000
choice=int(input("1.Deposit 2.Withdraw 3.Check Balance: "))

if choice==1:
    amt=int(input())
```

```
    balance+=amt
    print(balance)
elif choice==2:
    amt=int(input())
    if amt<=balance:
        balance-=amt
        print(balance)
    else:
        print("Insufficient")
elif choice==3:
    print(balance)
```

30⃣ Question:

Validate date (basic DD/MM/YYYY format).

```
d=int(input()); m=int(input()); y=int(input())

if m>=1 and m<=12:
    if d>=1 and d<=31:
        print("Valid Format (Basic Check)")
    else:
        print("Invalid Date")
else:
    print("Invalid Month")
```

🟢 LOGIC + NESTED

31⃣ Positive Even / Negative Odd etc.

```
n=int(input())
if n>0 and n%2==0:
    print("Positive Even")
elif n>0:
    print("Positive Odd")
elif n<0 and n%2==0:
    print("Negative Even")
else:
    print("Negative Odd")
```

32⃣ Middle Value of Three

```
a=int(input()); b=int(input()); c=int(input())
if (a>=b and a<=c) or (a<=b and a>=c):
    print(a)
elif (b>=a and b<=c) or (b<=a and b>=c):
    print(b)
else:
    print(c)
```

33⃣ Number in Multiple Ranges

```
n=int(input())
if n>=1 and n<=50:
```

```
    print("Range 1")
elif n>=51 and n<=100:
    print("Range 2")
else:
    print("Out of Range")
```

34 Time AM/PM

```
h=int(input())
if h<12:
    print("AM")
else:
    print("PM")
```

35 Rock Paper Scissors

```
u=input()
c="rock"

if u==c:
    print("Draw")
elif (u=="rock" and c=="scissors") or \
     (u=="paper" and c=="rock") or \
     (u=="scissors" and c=="paper"):
    print("User Wins")
else:
    print("Computer Wins")
```

36 Shopping Discount + Coupon

```
amount=int(input())
coupon=input()

if amount>=5000:
    discount=amount*0.2
elif amount>=2000:
    discount=amount*0.1
else:
    discount=0

if coupon=="SAVE100":
    discount+=100

print("Final:",amount-discount)
```

37 3 Login Attempts

```
correct="admin"
for i in range(3):
    pwd=input()
    if pwd==correct:
        print("Login Success")
        break
else:
```

```
    print("Account Locked")
```

38️⃣ Movie Ticket Pricing

```
age=int(input())
weekend=input()

if age<18:
    price=100
else:
    price=200

if weekend=="yes":
    price+=50

print(price)
```

39️⃣ Parking Fee

```
hours=int(input())
if hours<=2:
    fee=50
elif hours<=5:
    fee=100
else:
    fee=200
print(fee)
```

40️⃣ Income Tax

```
income=int(input())
if income<=250000:
    tax=0
elif income<=500000:
    tax=(income-250000)*0.05
else:
    tax=(250000*0.05)+(income-500000)*0.2
print(tax)
```

🧠 TRICKY & EDGE

41️⃣ Divisible by 2,3 both or none

```
n=int(input())
if n%2==0 and n%3==0:
    print("Both")
elif n%2==0:
    print("Divisible by 2")
elif n%3==0:
    print("Divisible by 3")
else:
    print("None")
```

42️⃣ Character Type Check

```
ch=input()
```

```
if ch>='A' and ch<='Z' or ch>='a' and ch<='z':
    print("Alphabet")
elif ch>='0' and ch<='9':
    print("Digit")
else:
    print("Special Character")
```

43 Harshad Number

```
n=int(input())
temp=n; s=0
while n>0:
    s+=n%10
    n//=10
if temp%s==0:
    print("Harshad")
else:
    print("Not Harshad")
```

44 Perfect Number

```
n=int(input())
s=0
for i in range(1,n):
    if n%i==0:
        s+=i
if s==n:
    print("Perfect Number")
else:
    print("Not Perfect")
```

45 Angles Form Triangle

```
a=int(input()); b=int(input()); c=int(input())
if a+b+c==180 and a>0 and b>0 and c>0:
    print("Valid")
else:
    print("Invalid")
```

46 Traffic Light

```
color=input()
if color=="red":
    print("Stop")
elif color=="yellow":
    print("Wait")
elif color=="green":
    print("Go")
```

47 Tolerance ±10%

```
value=int(input())
target=int(input())
low=target*0.9
high=target*1.1
```

```
if value>=low and value<=high:
    print("Within Tolerance")
else:
    print("Out of Range")
```

48 Username Validation

```
u=input()
if len(u)>=5:
    print("Valid")
else:
    print("Invalid")
```

49 Nested Distinction Rule

```
m=int(input())
if m>=40:
    if m>=75:
        print("Distinction")
    else:
        print("Pass")
else:
    print("Fail")
```

50 Business Rule Engine

```
amount=int(input())
member=input()

if member=="yes":
    if amount>5000:
        print("Gold Benefit")
    else:
        print("Silver Benefit")
else:
    print("No Membership Benefit")
```